

Solving Exponential and Logarithmic Equations

Matching bases, taking logarithms of both sides, converting a logarithmic equation to exponential form, and solving growth and decay models

Directions

- If both sides can be written as powers of the *same base*, set the exponents equal — that gives an exact answer with no logarithm at all.
 - If they cannot, take a logarithm of both sides and use $\log(M^p) = p \log M$ to bring the unknown down.
 - A logarithmic equation is solved by rewriting it in exponential form. Every answer must keep each logarithm's input positive.
 - Each modelling problem states its units. Give the answer in that unit, rounded as asked.
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1. Solve for x : $3^x = 81$

Answer: _____

2. Solve for x : $\log_4(x + 3) = 2$

Answer: _____

3. A culture starts with 500 bacteria and triples every hour, so the count after t hours is $N = 500 \cdot 3^t$. Find the number of bacteria after 4 hours.

Answer: _____

4. Solve $2^x = 20$, giving x to two decimal places.

Answer: _____

5. Solve for x : $5^{2x-1} = 125$

Answer: _____

6. An account opens with 2000 dollars and earns 6% interest compounded once a year, so its value after t years is $A = 2000(1.06)^t$ dollars. Find the value after 10 years, rounded to the nearest cent.

Answer: _____

7. Solve for x : $\log_2 x + \log_2(x - 2) = 3$

Answer: _____

8. A 100 milligram sample of a tracer decays so that the amount left after t days is $A = 100(0.5)^{t/12}$ milligrams. Find the number of days until 30 milligrams remain, rounded to two decimal places.

Answer: _____ days

9. Solve for x : $9^{x+1} = 27^x$

Answer: _____

10. Challenge. Consider $\log_5(2x + 3) - \log_5(x - 1) = 1$.

(a) Solve it. Give the exact value of x : _____

(b) Give that solution as a decimal, rounded to two places: _____

(c) State the condition the two logarithm inputs place on x , and explain why a candidate that breaks it would have to be thrown away.